Bangla LLM Hallucination Detection

Comprehensive & Beginner-Friendly Technical Architecture Guide

Primary Goal: Detect AI Lies in Bangla

Representations: BoW & TF-IDF

Classifiers: Naïve Bayes, LR, SVM

1. Introduction: What is Hallucination in AI?

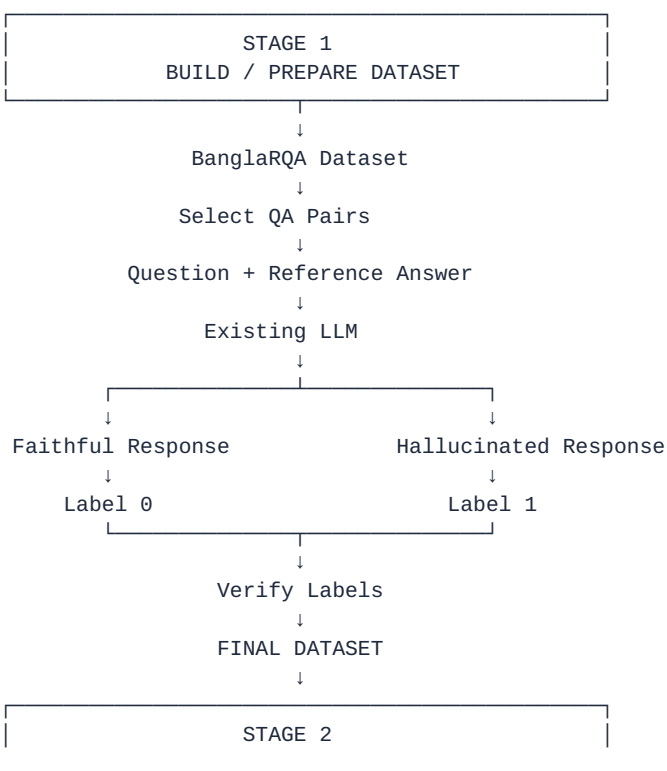
Artificial Intelligence tools like ChatGPT or Bangla Large Language Models (LLMs) can generate natural language responses. However, they often produce answers that sound extremely convincing but are **factually wrong or unsupported by the reference source text**. In Machine Learning research, these made-up responses are called **Hallucinations**.

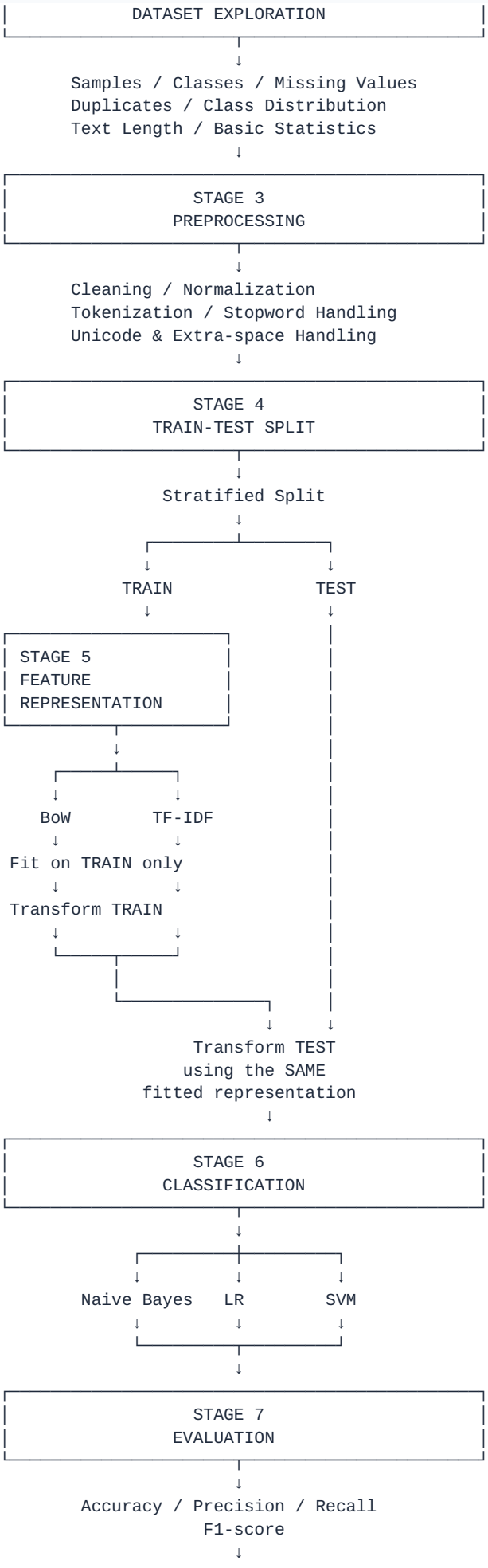
> 💡 **Real-World Analogy:**
> Imagine a student in an exam who forgets the correct answer to a history question. Instead of leaving it blank, they write a detailed, confident 3-page story that sounds smart but is completely made up. That confident, made-up story is a **hallucination**!

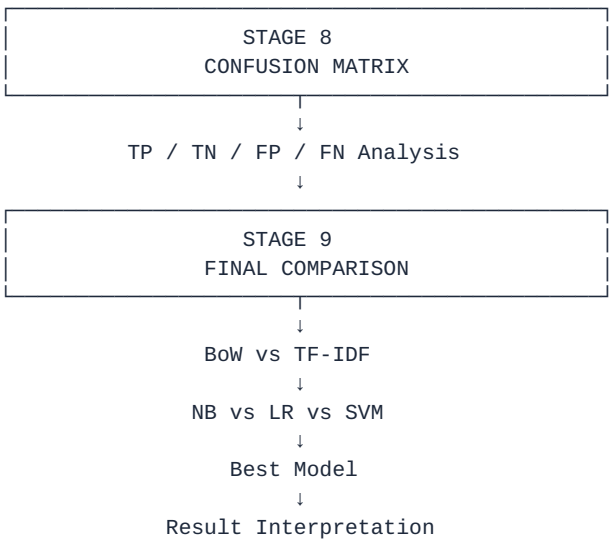
Project Goal: Build a classic Machine Learning pipeline that takes a Bangla Question, Reference Book/Passage, and AI Answer, and automatically predicts whether the AI response is Faithful (0) or Hallucinated (1).

2. Complete 9-Stage Pipeline Architecture

Below is the full high-level flowchart covering all 9 stages of our experimental workflow:




DATASET EXPLORATION
Samples / Classes / Missing Values
Duplicates / Class Distribution
Text Length / Basic Statistics
STAGE 3
PREPROCESSING
Cleaning / Normalization
Tokenization / Stopword Handling
Unicode & Extra-space Handling
STAGE 4
TRAIN-TEST SPLIT
Stratified Split
TRAIN
TEST
STAGE 5
FEATURE
REPRESENTATION
BoW
TF-IDF
Fit on TRAIN only
Transform TRAIN
Transform TEST
using the SAME
fitted representation
STAGE 6
CLASSIFICATION
Naive Bayes
LR
SVM
STAGE 7
EVALUATION
Accuracy / Precision / Recall
F1-score



3. Detailed Walkthrough of Each Stage

STAGE 1 — Build / Prepare the Final Dataset

We start with a standard benchmark dataset like **BanglaRQA**, select Question-Answer pairs, and use existing LLMs to generate responses. We then label each row:

- **Label 0 (Faithful):** The LLM answer accurately reflects the reference text.
- **Label 1 (Hallucinated):** The LLM answer contains false, unsupported, or fabricated details.

Example Dataset Structure:

Question	Reference Answer	LLM Answer	Label	Optional: Hallucination Type
বাংলাদেশের রাজধানী কোথায়?	ঢাকা বাংলাদেশের রাজধানী।	ঢাকা হলো বাংলাদেশের রাজধানী শহর।	**0** (Faithful)	None
রবীন্দ্রনাথ কবে নোবেল পান?	রবীন্দ্রনাথ ঠাকুর ১৯১৩ সালে নোবেল পান।	কাজী নজরুল ইসলাম ১৯১৩ সালে নোবেল পান।	**1** (Hallucinated)	Contradiction
কক্সবাজার কোথায় অবস্থিত?	কক্সবাজার বাংলাদেশে অবস্থিত।	কক্সবাজার পৃথিবীর দীর্ঘতম মিষ্টি পানির নদী।	**1** (Hallucinated)	Fabrication / Unsupported
কাজী নজরুল ইসলাম কে ছিলেন?	তিনি বাংলাদেশের জাতীয় কবি।	তিনি বাংলাদেশের জাতীয় কবি এবং বিখ্যাত বিজ্ঞানী।	**1** (Hallucinated)	Partial Hallucination

📌 **Hallucination Types Explained:**
- **Contradiction:** Directly opposes reference facts (e.g., wrong person or date).
- **Unsupported Information:** Adds claims that cannot be verified from reference text.

- **Fabrication:** Completely made-up fiction.
- **Partial Hallucination:** Starts correct, but sneaks in a false detail at the end.

STAGE 2 — Dataset Exploration (Exploratory Data Analysis)

Before running machine learning models, we analyze the raw dataset properties to ensure quality:

- **Total Samples:** e.g., 6,000 total QA pairs.
- **Class Balance:** e.g., 3,000 Faithful (50%) and 3,000 Hallucinated (50%).
- **Data Hygiene:** Check for missing values (NaNs), duplicate rows, or duplicate LLM responses.
- **Text Statistics:** Calculate average character/word lengths for Questions, References, and LLM Answers.

STAGE 3 — Text Preprocessing

Computers need standardized, clean text before converting words to numbers. We apply a sequential pipeline:

`Raw Text` ➔ `Cleaning` ➔ `Unicode Normalization` ➔ `Remove Unnecessary Characters/Symbols` ➔ `Remove Extra Spaces` ➔ `Tokenization` ➔ `Stopword Handling`

- **Unicode Normalization:** Standardizes Bangla font character encodings so the same word typed differently matches perfectly.
- **Tokenization:** Splitting sentences into individual words/tokens.
- **Stopword Handling:** Filtering out non-informative words (e.g., "এবং", "ও", "হয়", "আছে") that do not indicate factual truth.

STAGE 4 — Train-Test Split

We divide the cleaned dataset into two parts to evaluate real-world performance:

- **80% Training Set:** e.g., 4,800 samples used to teach vectorizers and classifiers.
- **20% Testing Set:** e.g., 1,200 samples kept completely hidden for final testing.

Stratified Split: We use stratification so both Train and Test sets maintain an exact 50/50 balance of Faithful and Hallucinated samples.

4. Understanding Data Leakage & Feature Representation (Stage 5)

> 🚨 **Critical Rule to Prevent Data Leakage:**
> Never fit Bag-of-Words (BoW) or TF-IDF on the combined dataset or test set! If the model learns word frequencies from test data beforehand, it is like giving a student the exam paper before test day.
>
> **Correct Workflow:**
> 1. Fit vocabulary / IDF **ONLY on Train data**.
> 2. Transform Train text into numbers.
> 3. Transform Test text using the **SAME fitted representation**.

Feature Representation Methods Explained Simply

1. **Bag of Words (BoW):** Counts the frequency of each word in a document. Simple and effective, but treats all words with equal weight.
2. **TF-IDF (Term Frequency - Inverse Document Frequency):** Gives higher numerical weight to important, rare words (e.g., "নোবেল", "কক্সবাজার") while lowering weights for common words that appear everywhere.

5. Classification, Evaluation & Confusion Matrix (Stages 6–8)

We run **6 total experiments** (2 Representations × 3 Classifiers):

- `BoW + Naïve Bayes` | `BoW + Logistic Regression` | `BoW + Support Vector Machine (SVM)`
- `TF-IDF + Naïve Bayes` | `TF-IDF + Logistic Regression` | `TF-IDF + Support Vector Machine (SVM)`

Evaluation Metrics & Why Recall Matters Most

Metric	Formula / Meaning	Importance for Hallucination Detection
Accuracy	(Correct Predictions) / (Total Predictions)	Good basic score, but can be misleading if classes are imbalanced.
Precision	True Positives / (True Positives + False Positives)	Measures how many flagged "Hallucinations" were actually false statements.
Recall	True Positives / (True Positives + False Negatives)	**CRITICAL!** Measures what percentage of actual AI lies were caught. Missing an AI lie is dangerous!
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	**The Primary Benchmark Metric.** Balances Precision and Recall into a single fair score.

Stage 8 — Confusion Matrix Layout

		Predicted Class	
		Predicted Faithful (0)	**Predicted Hallucinated (1)**
Actual Class	**Actual Faithful (0)**	**True Negative (TN)** Correctly identified true answer	**False Positive (FP)** Wrongly accused true answer of lying
	Actual Hallucinated (1)	**False Negative (FN)** ⚠️ Missed AI lie! (Dangerous)	**True Positive (TP)** Successfully caught AI lie!

6. Final Results Comparison & Model Selection (Stage 9)

At the end of the experiments, all results are compiled into a master comparison table to answer key research questions:

Feature Representation	Classifier Model	Accuracy	Precision	Recall	F1-Score
Bag of Words (BoW)	Naïve Bayes (NB)	—	—	—	—
Bag of Words (BoW)	Logistic Regression (LR)	—	—	—	—
Bag of Words (BoW)	Support Vector Machine (SVM)	—	—	—	—
TF-IDF	Naïve Bayes (NB)	—	—	—	—
TF-IDF	Logistic Regression (LR)	—	—	—	—
TF-IDF	Support Vector Machine (SVM)	—	—	—	—

Three Key Questions Answered in Interpretation:

1. **BoW vs TF-IDF:** Which feature representation captures Bangla context and hallucination markers more effectively?
2. **NB vs Logistic Regression vs SVM:** Which classifier handles text sparsity and decision boundaries best?
3. **Overall Best Model Selection:** Which combination achieves the highest F1-score? (e.g., If `TF-IDF + SVM` achieves the top F1-score, it will be crowned as our final benchmark model for Bangla LLM hallucination detection).

> 🌟 **Final Summary Takeaway:**
> By following this 9-stage pipeline, we systematically move from raw data collection to text normalization, enforce strict data leakage rules during train/test split, evaluate 6 baseline machine learning combinations, and identify the optimal model using F1-score and Recall.